MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) MODEL QUESTION PAPER

Semester V

MG5DSEBCA308 - Modern Web Application Development with React.js

(*2024 ADMISSION ONWARDS*)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks.

1. What is a Single Page Application (SPA)? [U] [CO1]
2. What is JSX? [U] [CO1]
3. A child component needs to change a piece of state that is maintained by its parent component. What React technique should be used to enable this? [A] [CO2]
4. What is conditional rendering in React? [U] [CO3]
5. What is the purpose of React Testing Library? [U] [CO4]

[2*5=10 Marks]

Part B

Short Answer Questions

Answer any 5 questions

Each question carries **6** marks.

6. Explain the component-based architecture of React.js. [U] [CO1]
7. A student wants to create a form containing Name, Email, and Age fields and display the entered values when the form is submitted. Explain how useState and controlled components can be applied to implement this form. [A] [CO2]
8. Explain how data can be fetched from an API in a React application using the Fetch API [U] [CO3]
9. Explain React Router and its role in navigation and routing in a React application. [U] [CO3]

10. Explain Context API and Redux Toolkit as approaches to state management in React. [U] [CO3]
11. Explain the main differences between REST and GraphQL. [U] [CO3]
12. Explain BEM naming convention. [U] [CO4]

[6*5=30 Marks]

Part C

Essay Questions

Answer any 2 questions

Each question carries **15** marks

13. Explain the process of setting up a React.js project [U] [CO1]
14. Explain the concept of React Hooks and the Rules of Hooks. [U] [CO2]
15. Explain the major techniques used for performance optimization in React. [U] [CO4]

[15*2=30 Marks]